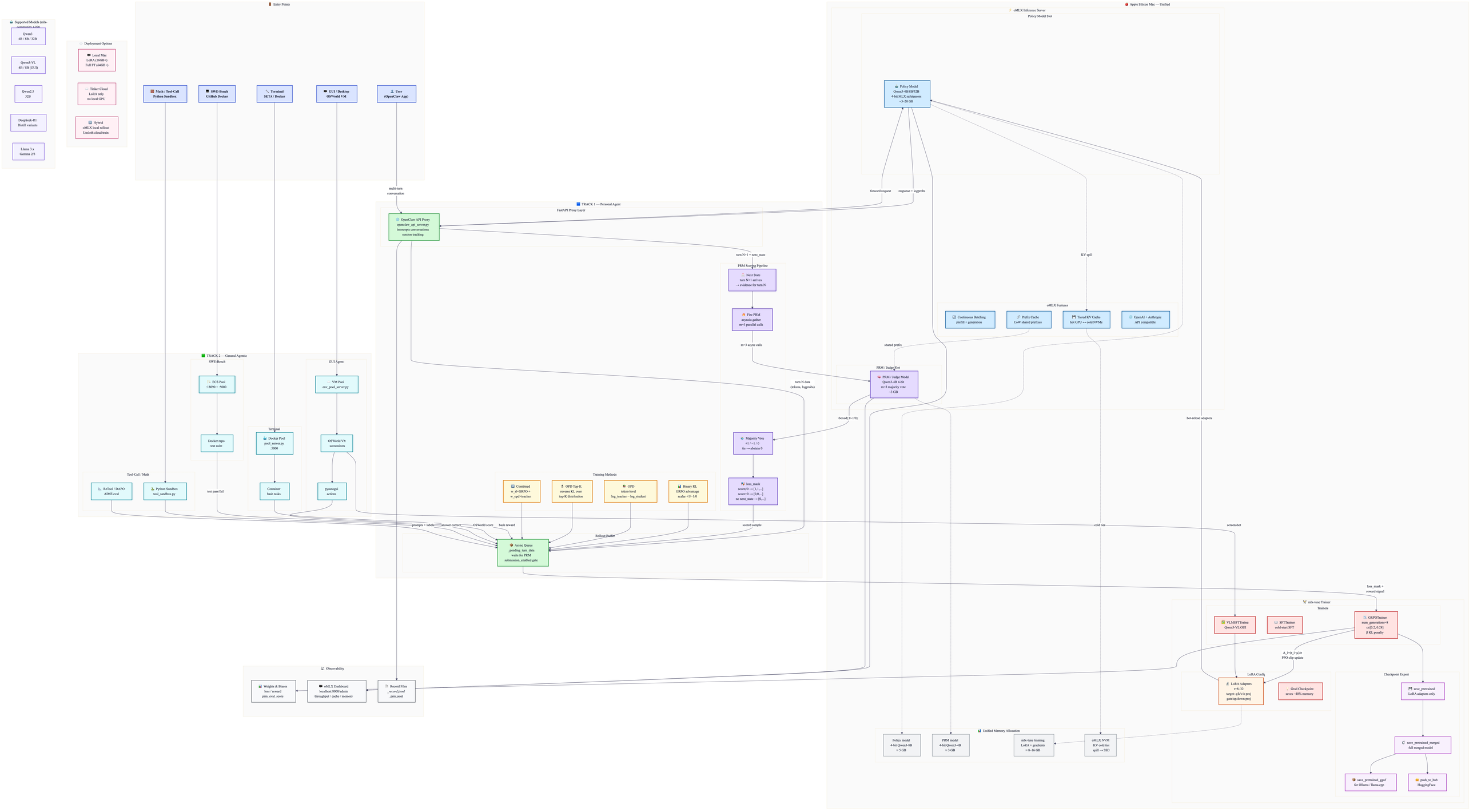

Supported Models (mlx-community 4-bit)
Qwen3 4B / 8B / 32B
Qwen3-VL 4B / 8B (GUI)
Qwen2.5 32B
DeepSeek-R1 Distill variants
Llama 3.x Gemma 2/3
Deployment Options
Local Mac LoRA (16GB+) Full FT (64GB+)
Tinker Cloud LoRA only no local GPU
Hybrid oMLX local rollout Unsloth cloud train
Entry Points
Math / Tool-Call Python Sandbox
SWE-Bench GitHub Docker
Terminal SETA / Docker
GUI / Desktop OSWorld VM
User (OpenClaw App)
multi-turn conversation
TRACK 1 — Personal Agent
FastAPI Proxy Layer
OpenClaw API Proxy openclaw_api_server.py intercepts conversations session tracking
turn N+1 = next_state
PRM Scoring Pipeline
Next State turn N+1 arrives → evidence for turn N
Fire PRM asyncio.gather m=3 parallel calls
m=3 async calls
turn N data (tokens, logprobs)
Majority Vote +1 / −1 / 0 tie → abstain 0
loss_mask score≠0 → [1,1,...] score=0 → [0,0,...] no next_state → [0,...]
scored sample
Training Methods
Combined w_rl×GRPO + w_opd×teacher
OPD Top-K reverse KL over top-K distribution
OPD token-level log_teacher − log_student
Binary RL GRPO advantage scalar +1/−1/0
Rollout Buffer
Async Queue _pending_turn_data waits for PRM submission_enabled gate
TRACK 2 — General Agentic
SWE-Bench
ECS Pool :18090 + :5000
Docker repo test suite
test pass/fail
Terminal
Docker Pool pool_server.py :5000
Container bash tasks
GUI Agent
VM Pool env_pool_server.py
OSWorld VM screenshots
pyautogui actions
Tool-Call / Math
ReTool / DAPO AIME eval
Python Sandbox tool_sandbox.py
prompts + labels
answer correct
OSWorld score
bash reward
Observability
Weights & Biases loss / reward prm_eval_score
oMLX Dashboard localhost:8000/admin throughput / cache / memory
Record Files _record.jsonl _prm.jsonl
Apple Silicon Mac — Unified
oMLX Inference Server
Policy Model Slot
Policy Model Qwen3-4B/8B/32B 4-bit MLX safetensors ~3–20 GB
forward request
response + logprobs
KV spill
oMLX Features
Continuous Batching prefill + generation
Prefix Cache CoW shared prefixes
Tiered KV Cache hot GPU ↔ cold NVMe
OpenAI + Anthropic API compatible
shared prefix
PRM / Judge Slot
PRM / Judge Model Qwen3-4B 4-bit m=3 majority vote ~3 GB
\boxed{1/-1/0}
hot-reload adapters
cold tier
screenshot
loss_mask + reward signal
mlx-tune Trainer
Trainers
VLMSFTTrainer Qwen3-VL GUI
SFTTrainer cold-start SFT
GRPOTrainer num_generations=8 ε∈[0.2, 0.28] β KL penalty
A_i=(r_i−μ)/σ PPO clip update
LoRA Config
LoRA Adapters r=8–32 target: q/k/v/o proj gate/up/down proj
Grad Checkpoint saves ~40% memory
Checkpoint Export
save_pretrained LoRA adapters only
save_pretrained_merged full merged model
save_pretrained_gguf for Ollama / llama.cpp
push_to_hub HuggingFace
Unified Memory Allocation
Policy model 4-bit Qwen3-8B ≈ 5 GB
PRM model 4-bit Qwen3-4B ≈ 3 GB
mlx-tune training LoRA + gradients ≈ 8–16 GB
oMLX NVM KV cold tier spill → SSD